

Naloxone: Drug information - UpToDate

Naloxone: Drug information - UpToDate

Official reprint from UpToDate®

www.uptodate.com © 2026 UpToDate, Inc. and/or its affiliates. All Rights Reserved.

Naloxone: Drug information

2026© UpToDate, Inc. and its affiliates and/or licensors. All Rights Reserved.

Contributor Disclosures

For additional information see "Naloxone: Patient drug information" and "Naloxone: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- FT Naloxone HCl [OTC];
- Kloxxado;
- Narcan [DSC];
- Narcan [OTC];
- Rextovy;
- Rezenopy;
- RiVive [OTC];
- Zimhi [DSC]

Brand Names: Canada

- Narcan;
- S.O.S. Naloxone Hydrochloride;
- TEVA Naloxone

Pharmacologic Category

- Antidote;
- Opioid Antagonist

Dosing: Adult

Note: Available routes of administration include IV (preferred), IM, SUBQ, and intranasal; other available routes (off-label) include inhalation via nebulization (adults only), intraosseous, and oral (adults only). Endotracheal administration is the least desirable and is supported by only anecdotal evidence (Ref).

Clonidine toxicity

Clonidine toxicity (off-label use): Limited data available: **Note:** Consultation with a clinical toxicologist or poison control center is highly recommended.

IV: Initial: 0.4 to 2 mg; may repeat doses as needed (Ref). Larger doses (up to 14 mg) have been reported (Ref).

Opioid reversal

Opioid reversal, life-threatening overdose:

Note: Patient selection: For patients without normal breathing but **with** a pulse (as assessed by a health care provider), the American Heart Association (AHA) recommends naloxone administration. For patients without normal breathing and **without** a pulse (as assessed by a health care provider), the AHA recommends initiation of CPR, the use of an automated external defibrillator, and consideration of naloxone administration (Ref). **Bystander administration:** Naloxone administration by bystanders (eg, law enforcement, family members, etc) has been shown to result in successful reversal and is encouraged for patients experiencing signs of opioid toxicity in the community (Ref); bystanders should seek immediate emergency medical assistance on behalf of the patient after administration of the first dose. **Larger doses:** Larger doses of naloxone may or may not be required to reverse the effects of illicitly manufactured fentanyl compounds and other high potency opioids; dosing decisions should be made based on patient-specific response (Ref).

IV (preferred):

Intermittent dosing:

Initial: **IV:** 0.04 to 2 mg; may repeat with escalating doses every 2 to 3 minutes, as needed (Ref); the need for ≥ 1 dose of naloxone is common (Ref). The initial dose of naloxone is empirical and based on patient- and overdose-specific factors (Ref). An initial dose at the higher end of the dosing range may be appropriate for patients with significant toxicity (eg, respiratory or cardiorespiratory arrest) (Ref); whereas, an initial dose on the lower end of the dosing range may be appropriate in patients with opioid dependence to avoid acute withdrawal or if there are concerns regarding concurrent stimulant overdose (Ref).

After reversal, may need to readminister dose(s) at a longer interval (ie, 20 to 60 minutes) depending on the type/duration of the opioid. If no response is observed after a cumulative dose of 10 mg, consider other causes of respiratory depression, including other drugs (Ref); however, relatively large doses of naloxone may be required to reverse the effects of illicitly manufactured fentanyl compounds and other high potency opioids (Ref).

Continuous infusion (off-label dosing): Note: Consider use with exposures to long-acting opioids (eg, methadone), extended-release products, or delayed absorption (eg, symptomatic patients who have concealed opioids internally via ingestion or vaginal/rectal insertion ["body packing" or "body stuffing"]) after initial naloxone response (Ref).

IV: Administer $\frac{2}{3}$ of the initial effective naloxone bolus per hour (usual dose: 0.25 to 6.25 mg/hour). For example, if the initial effective IV naloxone bolus is 1.5 mg, then the initial continuous IV infusion rate is 1 mg/hour. Consider administering $\frac{1}{2}$ of the initial effective bolus dose 15 minutes after initiation of the continuous infusion; adjust infusion rate, as needed, to assure adequate ventilation and prevent withdrawal symptoms (Ref). Alternatively, may calculate dosage/hour based on effective intermittent dose used and duration of adequate response seen (Ref).

IM, SUBQ: Note: May consider when IV route is not available.

Naloxone (0.4 mg/mL): **IM, SUBQ:** Initial: 0.04 to 2 mg; may repeat with escalating doses every 2 to 3 minutes, as needed (Ref); the need for ≥ 1 dose of naloxone is common (Ref). Seek immediate medical assistance after administration of the first dose. An initial dose at the higher end of the dosing range may be appropriate for patients with significant toxicity (eg, respiratory or cardiorespiratory arrest) (Ref); whereas, an initial dose on the lower end of the dosing range may be appropriate in patients with opioid dependence to avoid acute withdrawal or if there are concerns regarding concurrent stimulant overdose (Ref).

After reversal, may need to readminister dose(s) at a longer interval (ie, 20 to 60 minutes) depending on the type/duration of the opioid. If no response is observed after a cumulative dose of 10 mg total, consider other causes of respiratory depression, including other drugs (Ref); however, relatively large doses of naloxone may be required to reverse the effects of illicitly manufactured fentanyl compounds and other high potency opioids (Ref).

Naloxone (10 mg/0.4 mL auto-injector): **Note:** For self- or buddy-administration by military personnel or chemical incident responders; can be administered through clothing or MOPP4 personal protective

equipment.

Treatment: IM, SUBQ: 10 mg (contents of 1 auto-injector) as a single dose; may repeat as needed until emergency medical assistance becomes available. If the patient does not show some improvement after administering the dose, consider other causes of respiratory depression (eg, drugs of chemical terrorism/warfare agents). Seek immediate emergency medical assistance after administration of first dose.

Prevention: IM, SUBQ: 10 mg (contents of 1 auto-injector) as a single dose administered immediately prior to entering an area believed to be contaminated with high-potency opioids; may repeat if exposure to high-potency opioids is prolonged.

Naloxone (5 mg/0.5 mL prefilled syringe [Zimhi]): **IM, SUBQ:** 5 mg (contents of 1 prefilled syringe) as a single dose; may repeat every 2 to 3 minutes until emergency medical assistance becomes available. Seek immediate emergency medical assistance after administration of the first dose.

Inhalation via nebulization (off-label route): Note: May consider when IV route is not available. This administration method is not included in the AHA recommendations for initial management of an opioid-associated life-threatening emergency (Ref).

Nebulization: 2 mg; may repeat, as needed. Transition to IV or IM administration when possible (Ref).

Intranasal: Note: May consider when IV route is not available. Onset of action is slightly delayed and bioavailability reduced compared to IM or IV routes (Ref):

3, 4, or 8 mg (contents of 1 nasal spray) as a single dose in one nostril; may repeat with a new nasal spray every 2 to 3 minutes in alternating nostrils if patient does not respond or responds initially followed by recurrence of respiratory depression **or** 2 mg (1 mg per nostril; using injectable solution delivered with a mucosal atomization device); may repeat in 3 to 5 minutes if respiratory depression persists. Seek immediate medical assistance after administration of the first dose (Ref).

Endotracheal (off-label route): Note: May consider only when other routes are not available (Ref).

0.8 to 5 mg (ie, 2 to 2.5 times the initial recommended IV dose) (Ref).

Opioid reversal, acute therapeutic opioid use (eg, partial reversal in patients with respiratory depression): Note: Consider in patients receiving acute opioid therapy (eg, postoperative) who experience respiratory depression secondary to opioid therapy.

Intermittent dosing: IV: Initial: 0.05 to 0.2 mg every 2 to 3 minutes until desired response (eg, adequate ventilation and alertness without significant pain). Repeat doses may be needed within 1- to 2- hour intervals depending on type, dose, and timing of the last dose of opioid administered (Ref).

Continuous infusion (off-label dosing): Note: Consider use with exposures to long-acting opioids (eg, methadone), extended-release products, or delayed absorption (eg, symptomatic patients who have concealed opioids internally via ingestion or vaginal/rectal insertion ["body packing" or "body stuffing"]) after initial naloxone response (Ref).

IV: Administer $\frac{2}{3}$ of the initial effective naloxone bolus per hour. For example, if the initial effective IV naloxone bolus is 0.15 mg, then the initial continuous IV infusion rate is 0.1 mg/hour. Consider administering $\frac{1}{2}$ of the initial effective bolus dose 15 minutes after initiation of the continuous infusion; adjust infusion rate, as needed, to assure adequate ventilation and prevent withdrawal symptoms (Ref). Alternatively, may calculate dosage/hour based on effective intermittent dose used and duration of adequate response seen (Ref).

Opioid reversal, chronic therapeutic opioid use (eg, cancer pain) (off-label dosing): Note: Consider in patients receiving chronic opioid therapy (eg, treatment of cancer pain) who experience respiratory depression secondary to opioid therapy. Dilute 1 mL of a naloxone 0.4 mg/mL formulation with 9 mL of NS or SWFI for a total volume of 10 mL to achieve a 0.04 mg/mL concentration.

IV: 0.02 to 0.08 mg slow bolus; administer every 30 to 120 seconds until improvement in symptoms; if no response is observed after a cumulative naloxone dose of 1 mg, consider other causes of respi-

ratory depression. If respiratory depression is due to long-acting opioids, may consider administering naloxone as a continuous infusion starting at $\frac{2}{3}$ of the initial effective naloxone bolus per hour (or 0.2 mg per hour) (Ref).

Opioid-induced constipation

Opioid-induced constipation (alternative agent) (off-label use): **Note:** May be used in patients with laxative-refractory opioid-induced constipation; current guidelines recommend other peripherally acting mu-opioid receptor antagonists (eg, naldemedine, naloxegol) in this population (Ref). Optimal dose, frequency, and duration of therapy have not been established; efficacy data are limited (Ref).

Oral (administration of injectable formulation): 1 to 12 mg/day in 1 to 4 divided doses (Ref).

Opioid-induced pruritus

Opioid-induced pruritus (off-label use): Continuous IV infusion: 0.25 mcg/kg/hour (Ref). Doses up to ~3 mcg/kg/hour have been employed (Ref). However, doses >2 mcg/kg/hour are more likely to lead to reversal of analgesia and are not recommended (Ref). **Note:** Monitor pain control; verify that the naloxone is not reversing analgesia.

Opioid use disorder, naloxone challenge

Opioid use disorder, naloxone challenge (off-label use):

IM: 0.4 to 0.8 mg (Ref).

IV: 0.2 mg as a single dose. After 30 seconds, if withdrawal symptoms are present, stop the challenge and treat symptomatically; may repeat challenge in 24 hours. If no withdrawal symptoms and vital signs are stable, inject 0.6 mg and observe for 20 minutes. After 20 minutes, if withdrawal symptoms are present, stop the challenge and treat symptomatically; may repeat challenge in 24 hours. If no withdrawal symptoms are present may initiate naltrexone (Ref).

SUBQ: 0.8 mg as a single dose. After 20 minutes, if withdrawal symptoms are present, stop the challenge and treat symptomatically; may repeat challenge in 24 hours. If no withdrawal symptoms and vital signs are stable, may initiate naltrexone (Ref).

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

IV, IM, SUBQ, Inhalation, Intranasal, Oral, Endotracheal:

Altered kidney function:

eGFR ≥ 30 mL/minute/1.73 m²: No dosage adjustment necessary (metabolized in liver to inactive metabolites (Ref)) (Ref).

eGFR <30 mL/minute/1.73 m²: No dosage adjustment necessary (Ref). **Note:** Extended half-life and prolonged opioid antagonism was reported in a patient with advanced kidney impairment receiving naloxone by continuous infusion (Ref).

Hemodialysis, intermittent (thrice weekly): Unlikely to be significantly dialyzable (large V_d): No supplemental dose or dosage adjustment necessary (Ref). **Note:** Extended half-life and prolonged opioid antagonism was reported in a patient with advanced kidney impairment receiving naloxone by continuous infusion (Ref).

Peritoneal dialysis: Unlikely to be significantly dialyzable (large V_d): No dosage adjustment necessary (Ref). **Note:** Extended half-life and prolonged opioid antagonism was reported in a patient with advanced kidney impairment receiving naloxone by continuous infusion (Ref).

CRRT: No dosage adjustment likely to be necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment likely to be necessary (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Older Adult

Refer to adult dosing. The naloxone dose is the same in patients >50 years of age as younger patients (Ref).

Dosing: Pediatric

(For additional information see "Naloxone: Pediatric drug information")

Dosage guidance:

Dosage form information: IV administration results in the most rapid absorption; alternative routes may be used. Intranasal naloxone is as effective as IV/IM administration; intranasal dosage forms designed for adult use have been safely administered to pediatric patients for opioid intoxication/overdose in the prehospital setting; use of fixed-dose intranasal/IM/SUBQ formulations should be considered for pediatric patients in the prehospital setting (Ref).

Opioid intoxication/overdose

Opioid intoxication/overdose (full reversal):

IV (preferred), Intraosseous: Note: May be administered IM, intranasally, SUBQ, or endotracheally, but onset of action may be delayed, especially if patient has poor perfusion; doses may need to be repeated (Ref).

Infants and Children <5 years or ≤ 20 kg: IV, Intraosseous: 0.1 mg/kg/dose; repeat every 2 to 3 minutes if needed; monitor closely; may need to repeat doses (eg, every 20 to 60 minutes) if duration of action of opioid is longer than naloxone (Ref).

Children ≥ 5 years or >20 kg and Adolescents: IV, Intraosseous: 2 mg/dose; if no response, repeat every 2 to 3 minutes; monitor closely; may need to repeat doses (eg, every 20 to 60 minutes) if duration of action of opioid is longer than naloxone (Ref).

Endotracheal: Infants, Children, and Adolescents: Optimal endotracheal dose unknown; current expert recommendations are 2 to 3 times the IV dose (Ref).

IM, SUBQ: Note: IM or SUBQ absorption may be delayed or erratic.

Prefilled 5 mg syringe (Zimhi): Infants, Children, and Adolescents: IM, SUBQ: 5 mg (contents of 1 prefilled syringe) as a single dose; may repeat every 2 to 3 minutes if needed until emergency medical assistance becomes available. **Note:** Prefilled syringe is intended for children ≥ 12 years of age but may be used in infants and children <12 years of age if other dosage forms are not available (Ref).

Solutions for injection (eg, ampules, vials, prefilled 2 mg syringe): Infants, Children, and Adolescents: IM, SUBQ: 0.1 mg/kg/dose; maximum dose: 2 mg/dose; repeat every 2 to 3 minutes if needed; monitor closely; may need to repeat doses (eg, every 20 to 60 minutes) if duration of action of opioid is longer than naloxone (Ref).

Auto-injector (10 mg dose): Note: For self or buddy administration by military personnel or chemical incident responders.

Treatment: Children ≥ 12 years and Adolescents: IM, SUBQ: 10 mg (contents of 1 auto-injector) as a single dose; may repeat as needed until emergency medical assistance becomes available. If the patient does not show some improvement after administering the dose, consider other causes of respiratory depression. Seek immediate emergency medical assistance after administration of first dose.

Prevention: Children ≥ 12 years and Adolescents: IM, SUBQ: 10 mg (contents of 1 auto-injector) as a single dose administered immediately prior to entering an area believed to be contaminated with high potency opioids; may repeat if exposure to high potency opioids is prolonged.

Intranasal: Note: Onset of action is slightly delayed compared to IM or IV routes (Ref). In infants, the use of naloxone dosage forms that allow for weight-based dosing are preferred (ie, IV, intraosseous, endotracheal).

Intranasal formulations (eg, RiVive, Kloxxado, Rextovy, Narcan Nasal Spray): Infants, Children, and Adolescents: Intranasal: 3 mg, 4 mg, or 8 mg (contents of 1 nasal spray) as a single dose; may repeat every 2 to 3 minutes in alternating nostrils if needed until medical assistance becomes available (Ref).

Solution for injection (1 mg/mL injection) for intranasal administration: Adolescents ≥ 13 years: Intranasal: 2 mg (1 mg per nostril) (Ref). **Note:** Naloxone 0.4 mg/mL parenteral formulation has been evaluated and may be used for opioid overdose; however, due to the volume needed to administer the dose it is not ideal for nasal administration; should be used if intranasal administration is needed and a more concentrated naloxone is not readily available (Ref).

Continuous IV infusion: Limited data available: Infants, Children, and Adolescents: Continuous IV infusion: If continuous infusion is required after intermittent dosing, calculate the initial dosage/hour based on the effective intermittent dose used and duration of adequate response seen (Ref) **or** use two-thirds ($\frac{2}{3}$) of the initial effective naloxone bolus given as the hourly infusion (Ref); titrate dose as needed; 24 to 40 **mcg/kg/hour** has been reported (Ref). Doses as low as 2.5 **mcg/kg/hour** have been reported in adults and a dose of 160 **mcg/kg/hour** was reported in one neonate (Ref). **Note:** The infusion should be discontinued by reducing the infusion rate in decrements of 25%; closely monitor the patient (eg, pulse oximetry, respiratory rate) after each adjustment and after discontinuation of the infusion for recurrence of opioid-induced respiratory depression (Ref).

Reversal of respiratory depression from therapeutic opioid dosing

Reversal of respiratory depression from therapeutic opioid dosing:

Infants, Children, and Adolescents: IV, Intraosseous, IM, SUBQ: 0.001 to 0.005 mg/kg/dose; titrate to effect (Ref); AAP recommends a wider dosage range of 0.001 to 0.02 mg/kg/dose and may increase if necessary up to 0.1 mg/kg/dose (full reversal dose); maximum dose: 2 mg/dose; may repeat dose every 2 minutes as needed based on response; monitor patient closely; symptoms may recur if duration of action of opioid is longer than naloxone; repeat doses (eg, every 20 to 60 minutes) may be required (Ref).

Opioid-induced pruritus

Opioid-induced pruritus: Limited data available:

Prevention: Children ≥ 6 years and Adolescents ≤ 17 years: Continuous IV infusion: 0.25 **mcg/kg/hour** was used in a double-blind, prospective, randomized, placebo-controlled study (n=20) which showed lower incidence and severity of opioid-induced side effects (ie, pruritus, nausea) without a loss of pain control (Ref).

Treatment: Children ≥ 3 years and Adolescents: Continuous IV infusion: Initial: 2 **mcg/kg/hour**; if pruritus continues, may titrate by 0.5 **mcg/kg/hour** every few hours; dosing based on a retrospective study (n=30, age range: 3 to 20 years) with a reported mean ($\pm$ SD) dose of 2.3 ± 0.68 **mcg/kg/hour**; monitor closely; doses ≥ 3 **mcg/kg/hour** may increase risk for loss of pain control and patients may require an increase in opioid dose (Ref).

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Frequency not defined (any route of administration):

Cardiovascular: Flushing, hypertension, hypotension, presyncope, tachycardia, ventricular fibrillation, ventricular tachycardia

Dermatologic: Diaphoresis, piloerection, xeroderma

Endocrine & metabolic: Hot flash

Gastrointestinal: Abdominal cramps, abdominal pain, constipation, diarrhea, nausea, oral paresthesia, toothache, vomiting

Hepatic: Increased serum bilirubin

Local: Erythema at injection site, injection-site reaction

Nervous system: Agitation, asthenia, body pain, coma, confusion, disorientation, dizziness, encephalopathy, excessive crying (neonates), hallucination, headache, hyperreflexia (neonates), irritability, nervousness, paresthesia, restlessness, seizure (neonates), shivering, tonic-clonic seizure, tremor, withdrawal syndrome, yawning

Neuromuscular & skeletal: Muscle spasm, musculoskeletal pain

Respiratory: Dry nose, dyspnea, hypoxia, nasal congestion, nasal discomfort (pain), nasal mucosa swelling, respiratory depression, rhinitis, rhinorrhea, sneezing

Miscellaneous: Fever

Postmarketing (any route of administration):

Cardiovascular: Bradycardia

Nervous system: Aggressive behavior, anxiety, depression, drowsiness, loss of consciousness, malaise, outbursts of anger, pain, unresponsive to stimuli

Neuromuscular & skeletal: Back pain

Ophthalmic: Miosis

Respiratory: Pulmonary edema (Al-Azzawi 2021)

Contraindications

Hypersensitivity to naloxone or any component of the formulation

Warnings/Precautions

Concerns related to adverse effects:

- Acute opioid withdrawal: Administration of naloxone causes the release of catecholamines, which may precipitate acute withdrawal or unmask pain in those who regularly take opioids. Symptoms of acute withdrawal in opioid-dependent patients may include pain, tachycardia, hypertension, fever, sweating, abdominal cramps, diarrhea, nausea, vomiting, agitation, and irritability. In neonates born to mothers with opioid dependence, opioid withdrawal may be life-threatening and symptoms may include excessive crying, shrill cry, failure to feed, seizures, and hyperactive reflexes. In settings other than acute opioid overdose (eg, postoperative patients), carefully titrate the dose to reverse hypoventilation; do not fully awaken patient or reverse analgesic effect. Lower doses (eg, 2 mg nasal dose [off label]) are less likely to precipitate severe opioid withdrawal compared to higher doses (eg, 4 or 10 mg); however, the 2 mg dose may not provide an adequate and timely reversal in patients who have been exposed to an overdose of a potent or very high dose of opioids (Brenner 2021; Purssell 2021).

- Combativeness: Some patients may be agitated or combative when resuscitated with naloxone (Brenner 2021); there is greater risk of combativeness in patients using fentanyl (Gooley 2022).

Disease-related concerns:

- Cardiovascular disease: Use with caution in patients with cardiovascular disease or in patients receiving medications with potential adverse cardiovascular effects (eg, hypotension, pulmonary edema or arrhythmias); pulmonary edema and cardiovascular instability, including ventricular fibrillation, have been reported in association with abrupt reversal when using opioid antagonists.
- Seizures: Use caution in patients with history of seizures; avoid use in the treatment of meperidine-induced seizures.

Special populations:

- Pediatric: In opioid-dependent neonates, infants, children, and adolescents, administration of intranasal formulations of naloxone that are not weight-based may result in abrupt and complete reversal of opioid effects, leading to acute opioid withdrawal syndrome. It is preferable to utilize a naloxone formulation that allows for weight-based dosing when possible; titrate to effect.

Dosage form specific issues:

- Prefilled syringe: When administered to infants <1 year of age, monitor the injection site for residual needle parts and signs of infection.

Other warnings/precautions:

- Opioid overdose symptom recurrence: Recurrence of respiratory and/or CNS depression is possible if the opioid involved is long-acting. Continuously observe patients until there is no further risk of recurrent respiratory or CNS depression. Recurrence may be less likely following a heroin overdose as compared to nonheroin opioid overdoses; shorter observation times following naloxone administration after known heroin use may be warranted (Willman 2017). Time to recurrence may not be related to the naloxone initial dose (Wong 2019).
- Partial opioid agonist and mixed opioid agonist/antagonist overdose: Reversal of partial opioid agonists or mixed opioid agonist/antagonists (eg, buprenorphine, pentazocine) may be incomplete and larger or repeat doses of naloxone may be required.
- Postoperative reversal: Appropriate use: Excessive dosages should be avoided after use of opioids in surgery. Abrupt postoperative reversal may result in nausea, vomiting, sweating, tachycardia, hypertension, seizures, and other cardiovascular events (including pulmonary edema and arrhythmias).
- Substance use disorder involving opioid use: To prevent overdose deaths, there are initiatives to dispense naloxone for self- or buddy-administration to patients at risk of opioid overdose (eg, recipients of high-dose opioids, suspected or confirmed history of illicit opioid use) and individuals likely to be present in an overdose situation (eg, family members of illicit drug users) (Albert 2011; Bennett 2011). Clinical practice guidelines recommend patients being treated for opioid use disorder should be given prescriptions for naloxone. Patients and family members/significant others should be trained in the use of naloxone in overdose (ASAM 2020). Zimhi and intranasal products are indicated for emergency treatment. Needleless administration via nebulization and the intranasal route using the injectable solution (with a mucosal atomization device) by first responders and bystanders has also been described (Doe-Simkins 2009; Weber 2012). Needleless administration provides an alternative route of administration in patients with venous scarring due to IV injection of illicit drugs. There is a low incidence of death following naloxone reversal of opioid toxicity in patients who refuse transport to a health care facility (Wampler 2011). Nevertheless, patients who received naloxone in the out-of-hospital setting should seek immediate emergency medical assistance after the first dose due to the likelihood that respiratory and/or central nervous system depression will return.

Warnings: Additional Pediatric Considerations

In opioid-dependent neonates, infants, children, and adolescents, administration of intranasal formulations of naloxone that are not weight-based may result in abrupt and complete reversal of opioid effects, leading to acute opioid withdrawal syndrome. In neonates born to mothers with opioid dependence, opioid withdrawal may be life-threatening and symptoms may include excessive crying, shrill cry, failure to feed, seizures, and hyperactive reflexes. In settings other than acute opioid overdose (eg, postoperative patients), carefully titrate the dose to reverse hypoventilation; do not fully

awaken patient or reverse analgesic effect. Lower doses (eg, 2 mg nasal dose [off label]) are less likely to precipitate severe opioid withdrawal compared to higher doses (eg, 4 or 10 mg); however, the 2 mg dose may not provide an adequate and timely reversal in patients who have been exposed to an overdose of a potent or very high dose of opioids (Brenner 2021; Pursell 2021).

Product Availability

Rezenopy nasal spray: FDA approved April 2024; anticipated availability currently unknown. Information pertaining to this product within the monograph is pending revision. Consult the prescribing information for additional information.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Liquid, Nasal, as hydrochloride:

FT Naloxone HCl: 4 mg/0.1 mL (2 ea) [contains benzalkonium chloride, edetate (edta) disodium]

Kloxxado: 8 mg/0.1 mL (2 ea) [contains alcohol, usp, edetate (edta) disodium dihydrate, propylene glycol]

Narcan: 4 mg/0.1 mL (2 ea, 6 ea, 24 ea) [contains benzalkonium chloride, edetate (edta) disodium]

Rextovy: 4 mg/0.25 mL (2 ea)

Rezenopy: 10 mg/0.11 mL (2 ea) [latex free]

RiVive: 3 mg/0.1 mL (2 ea)

Generic: 4 mg/0.1 mL (1 ea, 2 ea, 24 ea)

Solution, Injection, as hydrochloride:

Generic: 0.4 mg/mL (1 mL); 4 mg/10 mL (10 mL)

Solution, Injection, as hydrochloride [preservative free]:

Generic: 0.4 mg/mL (1 mL)

Solution Cartridge, Injection, as hydrochloride:

Generic: 0.4 mg/mL (1 mL)

Solution Prefilled Syringe, Injection, as hydrochloride [preservative free]:

Zimhi: 5 mg/0.5 mL (0.5 mL [DSC]) [latex free]

Generic: 0.4 mg/mL (1 mL); 2 mg/2 mL (2 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Liquid (Kloxxado Nasal)

8MG/0.1ML (per each): \$75.00

Liquid (Naloxone HCl Nasal)

4MG/0.1ML (per each): \$48.00 - \$71.25

Liquid (Narcan Nasal)

4MG/0.1ML (per each): \$27.00

Liquid (Rextovy Nasal)

4 mg/0.25 mL (per each): \$27.00

Liquid (Rezenopy Nasal)

10MG/0.11ML (per each): \$90.00

Liquid (RiVive Nasal)

3MG/0.1ML (per each): \$18.00

Solution (Naloxone HCl Injection)

0.4 mg/mL (per mL): \$5.27 - \$23.72

4 mg/10 mL (per mL): \$6.60 - \$14.25

Solution Cartridge (Naloxone HCl Injection)

0.4 mg/mL (per mL): \$80.40

Solution Prefilled Syringe (Naloxone HCl Injection)

0.4 mg/mL (per mL): \$8.13 - \$8.40

2 mg/2 mL (per mL): \$9.00 - \$19.80

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Liquid, Nasal, as hydrochloride:

Narcan: 4 mg/0.1 mL (0.1 mL) [contains benzalkonium chloride, edetate (edta) disodium]

Generic: 4 mg/0.1 mL (0.1 mL)

Solution, Injection, as hydrochloride:

Generic: 0.4 mg/mL (1 mL, 10 mL); 1 mg/mL (2 mL)

Administration: Adult

IV push: Administer over 30 seconds as undiluted preparation **or** administer lower doses (eg, 0.02 to 0.04 mg) as a slow IV push using a diluted preparation (concentration 0.04 mg/mL).

IM, SUBQ: May administer IM or SUBQ if unable to obtain IV access.

Auto-injector: For IM or SUBQ use only. Intended for self- (10 mg/0.4 mL auto-injector only) or buddy administration; the person administering the medication should follow the printed instructions on the device or the electronic voice instructions coming from the speaker on the device (if applicable). In devices equipped with a voice instruction system, if the system does not operate properly, the device will still deliver the intended dose of naloxone when properly administered. Administer IM or SUBQ into the anterolateral aspect of the thigh; may be injected through clothing or MOPP4 personal protective equipment. Keep the 10 mg/0.4 mL auto-injector firmly in place for 5 seconds after injection (and after click and hiss sound). When being administered to infants <1 year of age, the thigh muscle should be pinched during administration. Following proper administration, a red indicator appears in the viewing window. The injection needle is not visible before, during, or after the injection. Patients who received naloxone in the out-of-hospital setting should seek immediate emergency

medical assistance after the first dose due to the likelihood that respiratory and/or CNS depression will return. Repeat doses may be required until emergency medical assistance becomes available or if prolonged exposure to high-potency opioids is expected (prophylaxis using the 10 mg/0.4 mL auto-injector only); a new device must be used as each device contains a single dose of naloxone.

Prefilled syringe (Zimhi): For IM or SUBQ use only. Periodically inspect solution through the viewing window during storage; replace with new device if solution is discolored yellow to brown, cloudy, or contains particles. Intended for buddy administration (by individuals ≥ 12 years of age); the person administering the medication should follow the printed instructions for use and the printed instructions on the device label. Place patient in supine position; administer IM or SUBQ into the anterolateral aspect of the thigh with the needle facing downwards; may be injected through clothing. When being administered to infants <1 year of age, the thigh muscle should be pinched during administration. Ensure the needle is embedded completely before injecting. After injection, slide the safety guard over the needle using one hand and return to blue case. Place patient in the lateral recumbent position (recovery position). Patients who received naloxone in the out-of-hospital setting should seek immediate emergency medical assistance after the first dose due to the likelihood that respiratory and/or CNS depression will return. Repeat doses may be required until emergency medical assistance becomes available. Used syringes should be given to health care provider for inspection and proper disposal.

Endotracheal (off-label route): Administer diluted preparation directly into endotracheal tube (Ref).

Inhalation via nebulization (off-label route): Administer diluted preparation (0.67 mg/mL) via nebulizer face mask (Ref).

Intranasal:

Administer initial dose as soon as possible. Do not prime or test the device prior to administration. Administer in alternating nostrils with each dose. Place the patient in the supine position and provide support to the back of the neck to allow the head to tilt back. Following administration, turn the patient on their side. Patients who received naloxone in the out-of-hospital setting should receive immediate emergency medical assistance after the first dose due to the likelihood that respiratory and/or CNS depression will return. Each container contains a single intranasal spray, do not reuse; if repeat administration is necessary a new container must be used.

Alternate intranasal administration instructions using generic injectable solution:

1 mg/mL formulation (preferred): Administer total dose equally divided into each nostril using a mucosal atomization device (MAD) (Ref).

0.4 mg/mL formulation: Administer total dose in repeated increments of 0.1 mL per nostril over ~ 2 minutes using a MAD (Ref). **Note:** Due to the volume needed to administer the dose, 0.4 mg/mL is not ideal for nasal administration but may be used if more concentrated naloxone is not readily available (Ref).

Note: If a MAD is not available, the solution may be sprayed into the nares without a MAD; however, a significant amount of drug may be lost likely due to swallowing and subsequent first-pass metabolism (Ref).

Oral (off-label route): For the management of opioid-induced constipation (off-label use), undiluted solution for injection may be administered orally (Ref).

Preparation for Administration: Adult

IV push (dilution for lower doses [eg, 0.02 to 0.04 mg]): Dilute 1 mL of a naloxone 0.4 mg/mL formulation with 9 mL of NS or SWFI for a total volume of 10 mL to achieve a concentration of 0.04 mg/mL (Ref).

IV infusion: Dilute naloxone 2 mg in 500 mL of NS or D5W to achieve a final concentration of 4 mcg/mL.

Inhalation via nebulization (off-label route): Dilute 2 mg of naloxone with 3 mL of NS to achieve a final concentration of 0.67 **mg/mL** (Ref).

Endotracheal (off-label route): Dilute recommended dose with 5 to 10 mL sterile water or NS.

Administration: Pediatric

Endotracheal: Dilute with NS prior to administration; follow with a flush ≥ 5 mL of NS and 5 consecutive positive-pressure ventilations (Ref).

Intranasal:

Intranasal formulation (eg, RiVive, Kloxxado, Rextovy, Narcan Nasal Spray): Administer initial dose as soon as possible. Do not prime or test the device prior to administration. Administer additional doses (when required) in alternating nostrils. Place the patient in the supine position and provide support to the back of the neck to allow the head to tilt back. Following administration, turn the patient on their side (recovery position). Each container contains a single intranasal spray; do not reuse; if repeat administration is necessary, a new container must be used.

Solution for injection for intranasal use: Note: If a mucosal atomizer device (MAD) is not available, the solution may be sprayed into the nares without a MAD; however, a significant amount of drug may be lost likely due to swallowing and subsequent first-pass metabolism (Ref).

1 mg/mL formulation (preferred): Administer total dose equally divided into each nostril using a MAD (Ref).

0.4 mg/mL formulation: Administer total dose in repeated increments of 0.1 mL per nostril over ~2 minutes using a MAD (Ref). **Note:** Due to the volume needed to administer the dose, 0.4 mg/mL is not ideal for nasal administration but may be used if more concentrated naloxone is not readily available (Ref).

Parenteral:

IV push: Administer over 30 seconds as undiluted preparation. May also be diluted and administer slow IV push (Ref).

Continuous IV infusion: Dilute in NS or D5W prior to administering as continuous IV infusion.

IM, Intraosseous, SUBQ: May administer IM, Intraosseous, or SUBQ if unable to obtain IV access.

Note: IM or SUBQ administration in pediatric patients, hypotensive patients, or patients with peripheral vasoconstriction or hypoperfusion may result in erratic or delayed absorption.

Dosage form specific: Periodically inspect solution through the viewing window during storage; replace with new device if solution is discolored yellow to brown, cloudy, or contains particles.

Prefilled 5 mg syringe (Zimhi): For IM or SUBQ use only. Intended for buddy administration (by individuals ≥ 12 years of age). The person administering the medication should follow the printed instructions for use and the printed instructions on the device label. Place patient in supine position; administer IM or SUBQ into the anterolateral aspect of the thigh with the needle facing downwards; may be injected through clothing. When being administered to infants < 1 year of age, the thigh muscle should be pinched during administration. Ensure the needle is embedded completely before injecting. After injection, slide the safety guard over the needle using one hand and return to case. Place patient in the lateral recumbent position (recovery position). Patients who received naloxone in the out-of-hospital setting should seek immediate emergency medical assistance after the first dose due to the likelihood that respiratory and/or CNS depression will return. Repeat doses may be required until emergency medical assistance becomes available. Used syringes should be given to health care provider for inspection and proper disposal.

Auto-injector (10 mg dose): Children ≥ 12 years and Adolescents: For IM or SUBQ use only. Intended for self or buddy administration by military personnel or chemical incident responders. The person administering the medication should follow the printed instructions on the device. Administer IM or SUBQ into the anterolateral aspect of the thigh; may be injected through clothing or MOPP4 personal protective equipment. Keep the 10 mg auto-injector firmly in place for 5 seconds after injection (and after click and hiss sound). Following proper administration, a red indicator appears in the viewing

window. The injection needle is not visible before, during, or after the injection. Immediately seek emergency medical assistance after the first dose due to the likelihood that respiratory and/or CNS depression will return. Repeat doses may be required until emergency medical assistance becomes available or if prolonged exposure to high-potency opioids is expected (prophylaxis using the 10 mg auto-injector only); a new device must be used for each dose as device only contains a single dose of naloxone.

Preparation for Administration: Pediatric

Endotracheal: Dilute to 1 to 2 mL with NS.

Parenteral:

IV push: May dilute naloxone 0.4 mg (1 mL ampule) with 9 mL of NS for a total volume of 10 mL to achieve a concentration of 0.04 mg/mL (Ref).

IV infusion: Dilute in NS or D5W to a final concentration of 4 **mcg**/mL (eg, naloxone 2 mg in 500 mL of fluid) per the manufacturer. ASHP recommends standard concentrations of 16 **mcg**/mL, 40 **mcg**/mL, or 400 **mcg**/mL for pediatric patients (Ref).

Storage/Stability

Solution, injection: Store at 20°C to 25°C (68°F to 77°F). Protect from light. Use IV infusion in NS or D5W within 24 hours of preparation.

Solution, auto-injector: Store at 15°C to 25°C (59°F to 77°F); excursions are permitted between 4°C and 40°C (39°F and 104°F); do not freeze. Store in the outer case provided; protect from heat. If the 10 mg/0.4 mL auto-injector was frozen, use another auto-injector (if available) and obtain emergency medical assistance immediately; the frozen 10 mg/0.4 mL auto-injector may be used after thawing.

Solution, prefilled syringe: Store at <25°C (<77°F); excursions are permitted up to 40°C (104°F); do not freeze. Store in the outer case provided; protect from light and extreme heat.

Solution, nasal spray:

Kloxxado: Store at 20°C to 25°C (68°F to 77°F); excursions are permitted between 5°C and 40°C (41°F and 104°F). Protect from light. Avoid freezing (freezes at –15°C [5°F]); however, if product is frozen, may thaw at room temperature for 15 minutes and still use.

Narcan (Rx product): Store at <25°C (<77°F); excursions are permitted up to 40°C (104°F). Protect from light. Avoid freezing; however, if product is frozen, may thaw at room temperature and still use.

Rextovy: Store at 20°C to 25°C (68°F to 77°F); excursions are permitted between 4°C and 40°C (39°F and 104°F). Protect from light. Avoid freezing (freezes at –15°C [5°F]); however, if product is frozen, may thaw at room temperature for 15 minutes and still use.

OTC:

3 mg per 0.1 mL product: Store at 20°C to 25°C (68°F to 77°F); avoid excessive heat above 40°C (104°F). Do not refrigerate.

4 mg per 0.1 mL products: Store at 2°C to 25°C (36°F to 77°F); avoid excessive heat above 40°C (104°F). Do not freeze; protect from light.

Pharmacy supply of emergency antidotes: Guidelines suggest that at least 20 to 40 mg of naloxone be stocked. Suggested amount is stated to be a sufficient quantity to treat 1 patient weighing 100 kg for an initial 8- to 24-hour period (Dart 2018); actual amount to be stocked should take into account site-specific and population-specific needs.

Use: Labeled Indications

Opioid reversal:

Naloxone (IV): For the complete or partial reversal of opioid toxicity (including respiratory depression) induced by natural and synthetic opioids (eg, propoxyphene, methadone, nalbuphine, butor-

phanol, pentazocine). Naloxone is also indicated for the diagnosis of suspected or known acute opioid overdose.

Naloxone (intranasal, IM, SUBQ [eg, Zimhi]): For the emergency treatment of known or suspected opioid overdose as manifested by respiratory and/or CNS depression. Intended for immediate administration as emergency therapy in settings where opioids may be present. Not a substitute for emergency medical care.

Naloxone 10 mg/0.4 mL autoinjector (IM, SUBQ): For use by military personnel and chemical incident responders for the treatment of respiratory and/or CNS depression in patients ≥ 12 years of age where use of high-potency opioids (eg, fentanyl analogues) as a chemical weapon is suspected. May also be used to prevent respiratory and/or CNS depression in military personnel and chemical incident responders entering an area contaminated with high-potency opioids.

Use: Off-Label: Adult

Clonidine toxicity; Opioid-induced constipation (oral administration); Opioid-induced pruritus; Opioid use disorder, naloxone challenge

Medication Safety Issues

Sound-alike/look-alike issues:

Naloxone may be confused with Lanoxin, nalbuphine, naltrexone

Naloxone (nasal sprays [Kloxxado, Narcan, Rextovy, ReVive]) may be confused with Netty (epinephrine nasal spray)

Narcan may be confused with Marcaine, Norcuron

Pediatric patients: High-risk medication:

KIDs List: Naloxone, when used in neonates for postpartum resuscitation, is identified on the Key Potentially Inappropriate Drugs in Pediatrics (KIDs) list; use should be avoided due to risk of seizure (strong recommendation; high quality of evidence) (PPA [McPherson 2025]).

International issues:

Narcan [multiple international markets] may be confused with Marcen brand name for ketazolam [Spain]

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Methylnaltrexone: May increase adverse/toxic effects of Opioid Antagonists. Specifically, the risk for opioid withdrawal may be increased. *Risk X: Avoid*

Naldemedine: Opioid Antagonists may increase adverse/toxic effects of Naldemedine. Specifically, the risk for opioid withdrawal may be increased. *Risk X: Avoid*

Naloxegol: Opioid Antagonists may increase adverse/toxic effects of Naloxegol. Specifically, the risk for opioid withdrawal may be increased. *Risk X: Avoid*

Pregnancy Considerations

Naloxone crosses the placenta.

Although naloxone may precipitate opioid withdrawal in the fetus in addition to the mother, treatment should not be withheld when needed in cases of maternal opioid overdose (ACOG 711 2017). When using the injection, starting at the low end of the dosing range is suggested to help avoid adverse fetal events but still provide treatment to the mother (Blandthorn 2018). Use of naloxone to test for opioid dependence during pregnancy is not recommended (ASAM 2020).

Naloxone is used off-label for the management of patients with opioid-induced pruritus, including women who received neuraxial opioids during labor and delivery (ACOG 209 2019; Kumar 2013; Miller 2011).

Breastfeeding Considerations

It is not known if naloxone is present in breast milk; however, systemic absorption following oral administration is minimal.

According to the manufacturer, the decision to continue or discontinue breastfeeding during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother.

Monitoring Parameters

Respiratory status (oxygenation and ventilation), level of consciousness, heart rate, blood pressure, temperature, signs or symptoms of opioid withdrawal.

Mechanism of Action

Pure opioid antagonist that competes and displaces opioids at opioid receptor sites

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Endotracheal, IM, SUBQ: 2 to 5 minutes; Inhalation via nebulization: ~5 minutes (Mycyk 2003); Intranasal: ~3 to 17 minutes (Barton 2005; Dietze 2019; Kelly 2005; Robertson 2009); IV: ~2 minutes.

Note: In clinical trials evaluating the use of naloxone in patients with an acute opioid overdose, onset of action is often defined as time to a predefined end point (eg, respiratory rate ≥ 10 ; Glasgow Coma Score ≥ 13 [Dietze 2019]) or is based on a predefined evaluation time point (eg, clinical response within 10 minutes of administration) (Kerr 2009).

Duration: ~30 to 120 minutes depending on route of administration; IV has a shorter duration of action than IM administration; since naloxone's action is shorter than that of most opioids, repeated doses are usually needed.

Absorption: Intranasal, IM, SUBQ: Pediatric patients: May be erratic or delayed.

Protein binding: Relatively weak (to albumin [major] and other plasma constituents).

Distribution: ~200 L (Fishman 1973).

Metabolism: Primarily hepatic via glucuronidation.

Bioavailability: 43% to 44% (0.4 mg IM compared to nasal [Narcan] dose); 37% (nasal [Kloxxado] compared to 2 mg IV dose); 40.7% (nasal [Rextovy] compared to 0.4 mg IV dose); $\leq 2\%$ (oral tablet compared to 1 mg IV infusion) (Smith 2012).

Half-life elimination: Neonates: Mean 3.1 ± 0.5 hours; Adults: IM, IV, or SUBQ: 0.5 to 1.5 hours; Intranasal: ~1.1 to 2 hours.

Time to peak: IM, Intranasal (Kloxxado), SUBQ: ~15 minutes; Intranasal (Narcan): 19.8 to 30 minutes.

Infants and children: Intranasal (Narcan): 10 minutes (Malmros Olsson 2021).

Excretion: Urine (as metabolites).

Patient Counseling Points

(For additional information see "Naloxone: Patient drug information")

What is this drug used for?

- It is used to avoid side effects from some drugs.
- It is used to treat some overdoses.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

Nose spray:

- Muscle pain
- Muscle spasm
- Headache
- Dry nose
- Runny or stuffy nose
- Swelling in the nose
- Constipation
- Tooth pain
- Dry skin
- Stomach pain
- Change in taste
- Dizziness
- Feeling tired or weak
- Sweating, shaking, upset stomach, or feeling angry when waking up

All injection products:

- Upset stomach or throwing up
- Dizziness
- Irritation where the shot is given
- Hot flashes
- Flushing

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Seizures
- Shortness of breath
- Chest pain or pressure, a fast heartbeat, or an abnormal heartbeat
- A burning, numbness, or tingling feeling that is not normal
- Mood changes
- Hallucinations (seeing or hearing things that are not there)

- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AT) Austria: Naloxon accord | Nyxoid;
- (AU) Australia: Naloxone | Naloxone Miniject | Prenoxad;
- (BE) Belgium: Narcan;
- (BR) Brazil: Narcan;
- (CH) Switzerland: Nyxoid;
- (CO) Colombia: Narcan;
- (CZ) Czech Republic: Nyxoid;
- (EE) Estonia: Nyxoid | Prenoxad;
- (ES) Spain: Nyxoid;
- (FI) Finland: Ventizolve;
- (FR) France: Nalone | Naloxone accord | Naloxone adapt pharma | Nalscue | Prenoxad | Ventizolve;
- (GB) United Kingdom: Ims naloxone | Naloxone | Nyxoid | Prenoxad;
- (IE) Ireland: Naloxone | Naloxone accord | Prenoxad | Ventizolve;
- (IT) Italy: Nyxoid;
- (JP) Japan: Naloxone hcl sankyo | Naloxone hydrochloride afp;
- (KR) Korea, Republic of: Exelan | Naloxone | Narone;
- (LT) Lithuania: Naloxonum;
- (LU) Luxembourg: Narcan | Nyxoid;
- (LV) Latvia: Naloxon | Narcan;
- (NL) Netherlands: Narcan;
- (NO) Norway: Naloxone accord | Nyxoid | Ventizolve;
- (NZ) New Zealand: Naloxone HCL | Nyxoid;
- (PR) Puerto Rico: Kloxxaao | Lifems naloxone | Naloxone HCL | Narcan | Rextovy | Zimhi;
- (PT) Portugal: Naloxona | Nyxoid;
- (QA) Qatar: N-Xone | Naxone | Nyxoid Spray;
- (SE) Sweden: Naloxone accord | Prenoxad | Respinal;
- (SI) Slovenia: Nyxoid;
- (TH) Thailand: DBL Naloxone | Naloxone | Narcan | Narcotan;
- (TW) Taiwan: Narcan

Use of UpToDate is subject to the Terms of Use.

REFERENCES

1. Abdelal R, Banerjee AR, Carlberg-Racich S, Darwaza N, Ito D, Epstein J. The need for multiple naloxone administrations for opioid overdose reversals: a review of the literature. *Subst Abus.* 2022a;43(1):774-784. doi:10.1080/08897077.2021.2010252 [PubMed 35112993]
2. Abdelal R, Raja Banerjee A, Carlberg-Racich S, et al. Real-world study of multiple naloxone administration for opioid overdose reversal among bystanders. *Harm Reduct J.* 2022b;19(1):49. doi:10.1186/s12954-022-00627-3 [PubMed 35596213]
3. Ahmed S, Bhivandkar S, Lonergan BB, Suzuki J. Microinduction of Buprenorphine/Naloxone: A Review of the Literature. *Am J Addict.* 2021;30(4):305-315. doi:10.1111/ajad.13135 [PubMed 33378137]
4. Ahmad SA, Scolnik D, Snehal V, Glatstein M. Use of naloxone for clonidine intoxication in the pediatric age group: case report and review of the literature. *Am J Ther.* 2015;22(1):e14-e16. doi:10.1097/MJT.0b013e318293b0e8 [PubMed 23782760]
5. Al-Azzawi M, Alshami A, Douedi S, Al-Taei M, Alsaoudi G, Costanzo E. Naloxone-induced acute pulmonary edema is dose-dependent: a case series. *Am J Case Rep.* 2021;22:e929412. doi:10.12659/AJCR.929412 [PubMed 33730013]
6. American College of Obstetricians and Gynecologists (ACOG). ACOG practice bulletin no. 209: obstetric analgesia and anesthesia. *Obstet Gynecol.* 2019;133(3):e208-e225. [PubMed 30801474]
7. American College of Obstetricians and Gynecologists (ACOG) Committee on Obstetric Practice. Committee opinion no. 711: opioid use and opioid use disorder in pregnancy. *Obstet Gynecol.* 2017;130(2):e81-e94. [PubMed 28742676]
8. American Society of Addiction Medicine (ASAM). The ASAM national practice guideline for the treatment of opioid use disorder: 2020 focused update. *J Addict Med.* 2020;14(2S)(suppl 1):1-91. doi:10.1097/ADM.0000000000000633 [PubMed 32511106]
9. American Society of Health-System Pharmacists (ASHP). Standardize 4 safety. <https://www.ashp.org/pharmapactice/standardize-4-safety-initiative>. Updated March 2025. Accessed April 8, 2026.
10. Albert S, Brason FW, Sanford CK, et al, "Project Lazarus: Community-Based Overdose Prevention in Rural North Carolina," *Pain Med*, 2011, 12(Suppl 2):77-85. [PubMed 21668761]
11. American Academy of Pediatrics (AAP) Committee on Drugs. Naloxone dosage and route of administration for infants and children: Addendum to emergency drug doses for infants and children. *Pediatrics.* 1990;86(3):484-485. [PubMed 2388800]
12. Bamshad MJ, Wasserman GS. Pediatric clonidine intoxications. *Vet Hum Toxicol.* 1990;32(3):220-223. [PubMed 2353431]
13. Bardsley R. Higher naloxone dosing may be required for opioid overdose. *Am J Health Syst Pharm.* 2019;76(22):1835-1837. doi:10.1093/ajhp/zxz208 [PubMed 31665765]
14. Barton ED, Colwell CB, Wilfe T, et al. Efficacy of intranasal naloxone as a needleless alternative for treatment of opioid overdose in the prehospital setting. *J Emerg Med.* 2005;29:265-271. doi:10.1016/j.jemermed.2005.03.007 [PubMed 16183444]
15. Bell A, Bennett AS, Jones TS, Doe-Simkins M, Williams LD. Amount of naloxone used to reverse opioid overdoses outside of medical practice in a city with increasing illicitly manufactured fentanyl in illicit drug supply. *Subst Abus.* 2019;40(1):52-55. doi:10.1080/08897077.2018.1449053 [PubMed 29558283]
16. Bennett AS, Bell A, Tomedi EG, et al, "Characteristics of an Overdose Prevention, Response, and Naloxone Distribution Program in Pittsburgh and Allegheny County, Pennsylvania," *J Urban Health*, 2011, 88(6):1020-30. [PubMed 21773877]

17. Bergasa NY, Alling DW, Talbot TL, et al, "Effects of Naloxone Infusions in Patients With the Pruritus of Cholestasis. A Double-Blind, Randomized, Controlled Trial," *Ann Intern Med*, 1995, 123(3):161-7. [PubMed 7598296]
18. Bisaga A, Mannelli P, Yu M, et al. Outpatient transition to extended-release injectable naltrexone for patients with opioid use disorder: a phase 3 randomized trial. *Drug Alcohol Depend*. 2018;187:171-178. doi:10.1016/j.drugalcdep.2018.02.023 [PubMed 29674251]
19. Blandthorn J, Bowman E, Leung L, Bonomo Y, Dietze P. Managing opioid overdose in pregnancy with take-home naloxone. *Aust N Z J Obstet Gynaecol*. 2018;58(4):460-462. [PubMed 29271100]
20. Boeuf B, Gauvin F, Guerguerian AM, et al, "Therapy of Shock With Naloxone: A Meta-Analysis," *Crit Care Med*, 1998, 26(11):1910-6. doi:10.1097/00003246-199811000-00038 [PubMed 9824088]
21. Boeuf B, Poirier V, Gauvin F, et al. Naloxone for shock. *Cochrane Database Syst Rev*. 2003;2003(4):CD004443. doi:10.1002/14651858.CD004443 [PubMed 14584016]
22. Boyer EW. Management of opioid analgesic overdose. *N Engl J Med*. 2012;367(2):146-155. doi:10.1056/NEJMr1202561 [PubMed 22784117]
23. Brenner DS, Stolbach AI, Zschoche J, Bright L. Severe prolonged agitation due to intranasal naloxone overexposure. *Am J Emerg Med*. 2021;42:261.e1-261.e2. doi:10.1016/j.ajem.2020.08.069 [PubMed 32919810]
24. Burket GA, Horowitz BZ, Hendrickson RG, Beauchamp GA. Endotracheal intubation in the pharmaceutical-poisoned patient: A narrative review of the literature. *J Med Toxicol*. 2021;17(1):61-69. doi:10.1007/s13181-020-00779-3 [PubMed 32394224]
25. Carpenter J, Murray BP, Atti S, Moran TP, Yancey A, Morgan B. Naloxone dosing after opioid overdose in the era of illicitly manufactured fentanyl. *J Med Toxicol*. 2020;16(1):41-48. doi:10.1007/s13181-019-00735-w [PubMed 31471760]
26. Chamberlain JM and Klein BL, "A Comprehensive Review of Naloxone for the Emergency Physician," *Am J Emerg Med*, 1994, 12(6):650-60. [PubMed 7945608]
27. Choi NG, Choi BY, DiNitto DM, Marti CN, Baker SD. Naloxone therapy for prescription and illicit opioid poisoning cases aged 50 + in the national poison data system, 2015-2020. *Clin Toxicol (Phila)*. 2022;60(4):499-506. doi:10.1080/15563650.2021.1981362 [PubMed 34554013]
28. Crockett SD, Greer KB, Heidelbaugh JJ, Falck-Ytter Y, Hanson BJ, Sultan S; American Gastroenterological Association Institute Clinical Guidelines Committee. American Gastroenterological Association Institute guideline on the medical management of opioid-induced constipation. *Gastroenterology*. 2019;156(1):218-226. doi:10.1053/j.gastro.2018.07.016 [PubMed 30340754]
29. Dart RC, Goldfrank LR, Erstad BL, et al. Expert consensus guidelines for stocking of antidotes in hospitals that provide emergency care. *Ann Emerg Med*. 2018;71(3):314-325.e1. doi:10.1016/j.annemergmed.2017.05.021 [PubMed 28669553]
30. Dellinger RP, Levy MM, Rhodes A, et al, "Surviving Sepsis Campaign: International Guidelines for Management of Severe Sepsis and Septic Shock, 2012," *Crit Care Med*, 2013, 41(2):580-637. [PubMed 23353941]
31. Dezfulian C, Orkin AM, Maron BA, et al. Opioid-associated out-of-hospital cardiac arrest: distinctive clinical features and implications for health care and public responses: A scientific statement from the American Heart Association. *Circulation*. 2021;143(16):e836-e870. doi:10.1161/CIR.0000000000000958 [PubMed 33682423]
32. Dietze P, Jauncey M, Salmon A, et al. Effect of intranasal vs intramuscular naloxone on opioid overdose: a randomized clinical trial. *JAMA Netw Open*. 2019;2(11):e1914977. doi:10.1001/jamanetworkopen.2019.14977 [PubMed 31722024]

33. Doe-Simkins M, Walley AY, Epstein A, et al, "Saved by the Nose: Bystander-Administered Intranasal Naloxone Hydrochloride for Opioid Overdose," *Am J Public Health*, 2009, 99(5):788-91. [PubMed 19363214]
34. Dowling J, Isbister GK, Kirkpatrick CM, Naidoo D, Graudins A. Population pharmacokinetics of intravenous, intramuscular, and intranasal naloxone in human volunteers. *Ther Drug Monit*. 2008;30(4):490-496. doi:10.1097/FTD.0b013e3181816214 [PubMed 18641540]
35. Dowell D, Haegerich TM, Chou R. CDC Guideline for prescribing opioids for chronic pain - United States, 2016. *MMWR Recomm Rep*. 2016;65(1):1-49. doi:10.15585/mmwr.rr6501e1 [PubMed 26987082]
36. Doyon S. Chapter 180. Opioids. In: Tintinalli J, Stapczynski JS, Ma OJ, et al. *Tintinalli's Emergency Medicine*. 7th ed. New York, NY: McGraw-Hill; 2010.
37. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.
38. Evans L, Rhodes A, Alhazzani W, et al. Surviving sepsis campaign: International guidelines for management of sepsis and septic shock 2021. *Crit Care Med*. 2021;49(11):e1063-e1143. doi:10.1097/CCM.0000000000005337 [PubMed 34605781]
39. Fishman J, Roffwarg H, Hellman L. Disposition of naloxone-7,8,3H in normal and narcotic-dependent men. *J Pharmacol Exp Ther*. 1973;187(3):575-580. [PubMed 4588968]
40. Gan TJ, Ginsberg B, Glass PS, et al, "Opioid-Sparing Effects of a Low-Dose Infusion of Naloxone in Patient-Administered Morphine Sulfate," *Anesthesiology*, 1997, 87(5):1075-81. [PubMed 9366459]
41. Gibson CM, Pass SE. Enteral naloxone for the treatment of opioid-induced constipation in the medical intensive care unit. *J Crit Care*. 2014;29(5):803-807. doi:10.1016/j.jcrc.2014.04.005 [PubMed 24866021]
42. Goldfrank L, Weisman RS, Errick JK, Lo MW. A dosing nomogram for continuous infusion intravenous naloxone. *Ann Emerg Med*. 1986;15(5):566-570. doi:10.1016/s0196-0644(86)80994-5 [PubMed 3963538]
43. Gooley B, Weston B, Colella MR, Farkas A. Outcomes of law enforcement officer administered naloxone. *Am J Emerg Med*. 2022;62:25-29. doi:10.1016/j.ajem.2022.09.041 [PubMed 36215781]
44. Gourlay GK, Coulthard K. The role of naloxone infusions in the treatment of overdoses of long half-life narcotic agonists: application to nor-methadone. *Br J Clin Pharmacol*. 1983;15(2):269-271. [PubMed 6849762]
45. Gugelmann HM, DeRoos FJ. Miscellaneous antihypertensives and pharmacologically related agents. In: MA, Lewin NA, Smith SW, Goldfrank LR, Hoffman RS, eds. *Goldfrank's Toxicologic Emergencies*. 11th ed. McGraw-Hill; 2019.
46. Hanes SD, Franklin M, Kuhl DA, Headley AS. Prolonged opioid antagonism with naloxone in chronic renal failure. *Pharmacotherapy*. 1999;19(7):897-901. doi:10.1592/phco.19.10.897.31557 [PubMed 10417042]
47. Hantson P, Evenepoel E, Ziade D, et al, "Adverse Cardiac Manifestations Following Dextro-propoxyphene Overdose: Can Naloxone Be Helpful?" *Ann Emerg Med*, 1995, 25(2):263-6. [PubMed 7832361]
48. Hoffman RS and Goldfrank LR, "The Poisoned Patient With Altered Consciousness. Controversies in the Use of a 'Coma Cocktail'," *JAMA*, 1995, 274(7):562-9. [PubMed 7629986]
49. Howlett C, Gonzalez R, Yerram P, Faley B. Use of naloxone for reversal of life-threatening opioid toxicity in cancer-related pain. *J Oncol Pharm Pract*. 2016;22(1):114-120. [PubMed 25227231] 10.1177/1078155214551589

50. Isbister GK, Heppell SP, Page CB, Ryan NM. Adult clonidine overdose: prolonged bradycardia and central nervous system depression, but not severe toxicity. *Clin Toxicol (Phila)*. 2017;55(3):187-192. doi:10.1080/15563650.2016.1277234 [PubMed 28107093]
51. Johnson C, Mayer P, and Grosz D, "Pulmonary Edema Following Naloxone Administration in a Healthy Orthopedic Patient," *J Clin Anesth*, 1995, 7(4):356-7. [PubMed 7546765]
52. Kaplan JL, Marx JA, Calabro JJ, et al, "Double-Blind, Randomized Study of Nalmefene and Naloxone in Emergency Department Patients With Suspected Narcotic Overdose," *Ann Emerg Med*, 1999, 34(1):42-50. [PubMed 10381993]
53. Kattwinkel J, Perlman JM, Aziz K, et al. Part 15: neonatal resuscitation: 2010 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2010;122(18)(suppl 3):909-919. [PubMed 20956231]
54. Kelly A, Kerr D, Dietze P, et al. Randomised trial of intranasal versus intramuscular naloxone in prehospital treatment for suspected opioid overdose. *Med J Aust*. 2005;182(1):24-27. doi:10.5694/j.1326-5377.2005.tb06550.x [PubMed 15651944]
55. Kendrick WD, Woods AM, Daly MY, et al, "Naloxone Versus Nalbuphine Infusion for Prophylaxis of Epidural Morphine-Induced Pruritus," *Anesth Analg*, 1996, 82(3):641-7. [PubMed 8623975]
56. Kerr D, Dietze P, Kelly AM. Intranasal naloxone for the treatment of suspected heroin overdose. *Addiction*. 2008;103(3):379-386. doi:10.1111/j.1360-0443.2007.02097.x [PubMed 18269360]
57. Kerr D, Kelly AM, Dietze P, Jolley D, Barger B. Randomized controlled trial comparing the effectiveness and safety of intranasal and intramuscular naloxone for the treatment of suspected heroin overdose. *Addiction*. 2009;104(12):2067-2074. doi:10.1111/j.1360-0443.2009.02724.x [PubMed 19922572]
58. Kjellberg F and Tramer MR, "Pharmacological Control of Opioid-Induced Pruritus: A Quantitative Systematic Review of Randomized Trials," *Eur J Anaesthesiol*, 2001, 18(6):346-57. [PubMed 11412287]
59. Kleinman ME, Chameides L, Schexnayder SM, et al. Part 14: Pediatric advanced life support: 2010 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2010; 122(18 Suppl 3):876-908. [PubMed 20956230]
60. Kloxxaдо (naloxone) [prescribing information]. Columbus, OH: Hikma Specialty USA Inc; September 2025.
61. Kumar K, Singh SI. Neuraxial opioid-induced pruritus: an update. *J Anaesthesiol Clin Pharmacol*. 2013;29(3):303-307. [PubMed 24106351]
62. Lavonas EJ, Dezfulian C. Impact of the opioid epidemic. *Crit Care Clin*. 2020;36(4):753-769. doi:10.1016/j.ccc.2020.07.006 [PubMed 32892827]
63. Lavonas EJ, Drennan IR, Gabrielli A, et al. Part 10: special circumstances of resuscitation: 2015 American Heart Association guidelines update for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2015;132(18)(suppl 2):S501-S518. doi:10.1161/CIR.0000000000000264 [PubMed 26472998]
64. Lebin JA, Chen BC, Valento MJ. Reversal of pediatric opioid toxicity with take-home naloxone: a case report. *J Med Toxicol*. 2019;15(2):134-135. doi:10.1007/s13181-018-0695-z [PubMed 30632074]
65. Lewis JM, Klein-Schwartz W, Benson BE, et al. Continuous naloxone infusion in pediatric narcotic overdose. *Am J Dis Child*. 1984;138(10):944-946. [PubMed 6475855]
66. Liu M, Wittbrodt E. Low-dose oral naloxone reverses opioid-induced constipation and analgesia. *J Pain Symptom Manage*. 2002;23(1):48-53. doi:10.1016/s0885-3924(01)00369-4 [PubMed 11779668]

67. Malmros Olsson E, Lönnqvist PA, Stiller CO, Eksborg S, Lundeberg S. Rapid systemic uptake of naloxone after intranasal administration in children. *Paediatr Anaesth*. 2021;31(6):631-636. doi:10.1111/pan.14175 [PubMed 33687794]
68. Maxwell LG, Kaufmann SC, and Bitzer S. The effects of a small-dose naloxone infusion on opioid-induced side effects and analgesia in children and adolescents treated with intravenous patient-controlled analgesia: A double-blind, prospective, randomized, controlled study. *Anesth Analg*. 2005;100(4):953-958. [PubMed 15781505]
69. McPherson C, Meyers RS, Thackray J, et al. Pediatric Pharmacy Association 2025 KIDs List of key potentially inappropriate drugs in pediatrics. *J Pediatr Pharmacol Ther*. 2025;30(4):422-439. doi:10.5863/JPPT-25-00061 [PubMed 40821415]
70. Merchan C, Altshuler D, Papadopoulos J. Methylnaltrexone versus naloxone for opioid-induced constipation in the medical intensive care unit. *Ann Pharmacother*. 2017;51(3):203-208. doi:10.1177/1060028016677310 [PubMed 28168885]
71. Merigian KS, "Cocaine-Induced Ventricular Arrhythmias and Rapid Atrial Fibrillation Temporally Related to Naloxone Administration," *Am J Emerg Med*, 1993, 11(1):96-7. [PubMed 8447884]
72. Miller JL, Hagemann TM. Use of pure opioid antagonists for management of opioid-induced pruritus. *Am J Health Syst Pharm*. 2011;68(15):1419-1425. [PubMed 21785031]
73. Mokhlesi B, Leikin JB, Murray P, Corbridge TC. Adult toxicology in critical care: Part II: specific poisonings. *Chest*. 2003;123(3):897-922. doi:10.1378/chest.123.3.897 [PubMed 12628894]
74. Moss RB, Carlo DJ. Higher doses of naloxone are needed in the synthetic opioid era. *Subst Abuse Treat Prev Policy*. 2019;14(1):6. doi:10.1186/s13011-019-0195-4 [PubMed 30777088]
75. Müller-Lissner S, Bassotti G, Coffin B, et al. Opioid-induced constipation and bowel dysfunction: a clinical guideline. *Pain Med*. 2017;18(10):1837-1863. doi:10.1093/pm/pnw255 [PubMed 28034973]
76. Mycyk MB, Szyszko AL, and Aks SE, "Nebulized Naloxone Gently and Effectively Reverses Methadone Intoxication," *J Emerg Med*, 2003, 24(2):185-7. [PubMed 12609650]
77. Naloxone hydrochloride [prescribing information] 0.4 mg/mL Carpuject syringe. Lake Forest, IL: Hospira Inc; May 2019.
78. Naloxone hydrochloride [prescribing information] 0.4 mg/mL single-dose and multi-dose vials. Lake Forest, IL: Hospira Inc; August 2019.
79. Naloxone hydrochloride [prescribing information] 2 mg/2 mL prefilled syringe. Deerfield, IL: Baxter Healthcare Corporation; October 2024.
80. Naloxone hydrochloride [prescribing information] 2 mg/2 mL prefilled syringe. Lake Zurich, IL: Frensius Kabi; May 2023.
81. Naloxone hydrochloride [prescribing information] 2 mg/2 mL prefilled syringe. Princeton, NJ: Dr. Reddy's Laboratories, Inc; July 2024.
82. Naloxone hydrochloride [prescribing information] 10 mg/0.4 mL auto-injector. Richmond, VA: Kaleo Inc; February 2022.
83. Naloxone hydrochloride [prescribing information]. Berkeley Heights, NJ: Hikma Pharmaceuticals USA Inc; May 2024.
84. Naloxone hydrochloride [prescribing information]. Lake Forest, IL: Akorn, Inc; July 2019.
85. Naloxone hydrochloride [prescribing information]. Lake Forest, IL: Hospira, Inc; July 2023.
86. Naloxone hydrochloride [prescribing information]. Malvern, PA: Endo USA; March 2024.
87. Naloxone hydrochloride nasal spray [prescribing information]. El Monte, CA: International Medications Systems, Limited; March 2023.

88. Narcan Nasal Spray (naloxone hydrochloride) [prescribing information]. Plymouth Meeting, PA: Emergent Devices Inc; received November 2025.
89. Narcan (naloxone hydrochloride) nasal spray OTC [prescribing information]. Plymouth Meeting, PA: Emergent Devices Inc; received August 2023.
90. Nelson LS, Howland MA, Lewin NA, et al. Opioid antagonists. *Goldfrank's Toxicologic Emergencies*. 11th ed. McGraw-Hill; 2019.
91. Neumar RW, Otto CW, Link MS, et al. Part 8: adult advanced cardiovascular life support: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation*. 2010;122(18 suppl 3):S729-S767. doi:10.1161/CIRCULATIONAHA.110.970988 [PubMed 20956224]
92. O'Connor PG and Kosten TR, "Rapid and Ultrarapid Opioid Detoxification Techniques," *JAMA*, 1998, 279(3):229-34. [PubMed 9438745]
93. Olsen KS, "Naloxone Administration and Laryngospasm Followed by Pulmonary Edema," *Intensive Care Med*, 1990, 16(5):340-1. [PubMed 2212264]
94. O'Malley-Dafner L, Davies P, "Naloxone-Induced Pulmonary Edema," *Am J Nurs*, 2000, 100(11):24AA-JJ.
95. Panchal AR, Bartos JA, Cabañas JG, et al; Adult Basic and Advanced Life Support Writing Group. Part 3: adult basic and advanced life support: 2020 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2020;142(16)(suppl 2):S366-S468. doi:10.1161/CIR.0000000000000916 [PubMed 33081529]
96. Pasero C and McCaffery M, "Reversing Respiratory Depression With Naloxone," *Am J Nurs*, 2000, 100(2):26. [PubMed 10683639]
97. Perry HE and Shannon MW. Diagnosis and management of opioid- and benzodiazepine-induced comatose overdose in children. *Curr Opin Pediatr*. 1996;8(3):243-247. [PubMed 8814402]
98. Phan NQ, Bernhard JD, Luger TA, Ständer S. Antipruritic treatment with systemic μ -opioid receptor antagonists: a review. *J Am Acad Derm*. 2010;63(4):680-688. [PubMed 20462660]
99. Purssell R, Godwin J, Moe J, et al. Comparison of rates of opioid withdrawal symptoms and reversal of opioid toxicity in patients treated with two naloxone dosing regimens: a retrospective cohort study. *Clin Toxicol (Phila)*. 2021;59(1):38-46. doi:10.1080/15563650.2020.1758325 [PubMed 32401548]
100. Refer to manufacturer's labeling.
101. Rextovy (naloxone) [prescribing information]. El Monte, CA: International Medication Systems; May 2024.
102. Rezenopy (naloxone) [prescribing information]. Lexington, KY: Summit Biosciences; April 2024.
103. RiVive (naloxone) [prescribing information]. Bethesda, MD: Harm Reduction Therapeutics Inc; received July 2023.
104. Robertson TM, Hendey GW, Stroh G, Shalit M. Intranasal naloxone is a viable alternative to intravenous naloxone for prehospital narcotic overdose. *Prehosp Emerg Care*. 2009;13(4):512-515. doi:10.1080/10903120903144866 [PubMed 19731165]
105. Robinson A, Wermeling DP. Intranasal naloxone administration for treatment of opioid overdose. *Am J Health Syst Pharm*. 2014;71(24):2129-2135. doi:10.2146/ajhp130798 [PubMed 25465584]
106. Salvucci AA Jr, Eckstein M, and Iscovich AL, "Submental Injection of Naloxone," *Ann Emerg Med*, 1995, 25(5):719-20. [PubMed 7741359]
107. Sandelich S, Hooley G, Hsu G, et al. Acute opioid overdose in pediatric patients. *J Am Coll Emerg Physicians Open*. 2024;5(2):e13134. doi:10.1002/emp2.13134 [PubMed 38464332]

108. Seger DL, Loden JK. Naloxone reversal of clonidine toxicity: dose, dose, dose. *Clin Toxicol (Phila)*. 2018;56(10):873-879. doi:10.1080/15563650.2018.1450986 [PubMed 29544366]
109. Sheno RP, Timm N; Committee on Drugs; Committee on Pediatric Emergency Medicine. Drugs used to treat pediatric emergencies. *Pediatrics*. 2020;145(1):e20193450. doi:10.1542/peds.2019-3450 [PubMed 31871244]
110. Smith K, Hopp M, Mundin G, et al. Low absolute bioavailability of oral naloxone in healthy subjects. *Int J Clin Pharmacol Ther*. 2012;50(5):360-367. doi:10.5414/cp201646 [PubMed 22541841]
111. Soyka M, Kranzler HR, van den Brink W, Krystal J, Möller HJ, Kasper S; WFSBP Task Force on Treatment, Guidelines for Substance Use Disorders. The World Federation of Societies of Biological Psychiatry (WFSBP) guidelines for the biological treatment of substance use and related disorders. Part 2: opioid dependence. *World J Biol Psychiatry*. 2011;12(3):160-187. doi:10.3109/15622975.2011.561872 [PubMed 21486104]
112. Storrow AB, Wians FH Jr, Mikkelsen SL, et al, "Does Naloxone Cause a Positive Urine Opiate Screen?" *Ann Emerg Med*, 1994, 24(6):1151-3. [PubMed 7978599]
113. Substance Abuse and Mental Health Services Administration (SAMHSA). Medications for Opioid Use Disorder. Treatment Improvement Protocol (TIP) Series 63. Publication No. PEP21-02-01-002. Substance Abuse and Mental Health Services Administration; 2021. <https://store.samhsa.gov/product/TIP-63-Medications-for-Opioid-Use-Disorder-Full-Documents/PEP21-02-01-002>. Accessed February 7, 2022.
114. Suzuki J, Mittal L, Woo SB. Sublingual buprenorphine and dental problems: a case series. *Prim Care Companion CNS Disord*. 2013;15(5):PCC.13101533. doi:10.4088/PCC.13101533 [PubMed 24511440]
115. Swift A, Wilson M. Reversal of the effects of clonidine using naloxone. *Anaesth Rep*. 2019;7(1):4-6. doi:10.1002/anr3.12004 [PubMed 32051935]
116. Tandberg D, Abercrombie D. Treatment of heroin overdose with endotracheal naloxone. *Ann Emerg Med*. 1982;11(8):443-445. doi:10.1016/s0196-0644(82)80045-0 [PubMed 7103164]
117. Tenenbein M. Continuous naloxone infusion for opiate poisoning in infancy. *J Pediatr*. 1984;105(4):645-648. doi:10.1016/s0022-3476(84)80440-0 [PubMed 6481543]
118. Terranella A, Guy G Jr, Mikosz C. Naloxone dispensing to youth ages 10-19: 2017-2022. *Pediatrics*. 2024;154(4):e2023065137. doi:10.1542/peds.2023-065137 [PubMed 39262344]
119. Toce MS, Freiman E, O'Donnell KA, Burns MM. Clinical effects of pediatric clonidine exposure: A retrospective cohort study at a single tertiary care center. *J Emerg Med*. 2021;60(1):58-66. doi:10.1016/j.jemermed.2020.09.018 [PubMed 33036823]
120. Trujillo MH, Guerrero J, Fragachan C, et al, "Pharmacologic Antidotes in Critical Care Medicine: A Practical Guide for Drug Administration," *Crit Care Med*, 1998, 26(2):377-91. [PubMed 9468179]
121. Vahabzadeh M, Banagozar Mohammadi A. Heroin body-packing and naloxone. *Lancet*. 2019;393(10177):e35. doi:10.1016/S0140-6736(19)30502-1 [PubMed 30910307]
122. van Lemmen M, Florian J, Li Z, et al. Opioid overdose: Limitations in naloxone reversal of respiratory depression and prevention of cardiac arrest. *Anesthesiology*. 2023;139(3):342-353. doi:10.1097/ALN.0000000000004622 [PubMed 37402248]
123. Van Vonderen JJ, Siew ML, Hooper SB, et al, "Effects of Naloxone on the Breathing Pattern of a Newborn Exposed to Maternal Opiates," *Acta Pediatr*, 2012, 101(7):e309-12. [PubMed 22372574]
124. Vanden Hoek TL, Morrison LJ, Shuster M, et al. Part 12: cardiac arrest in special situations: 2010 American Heart Association guidelines for cardiopulmonary resuscitation and Emergency cardiovascular care. *Circulation*. 2010;122(18 suppl 3):S829-S861. doi:10.1161/CIRCULATIONAHA.110.971069 [PubMed 20956228]

125. Vrchoticky T, "Naloxone for the Treatment of Narcotic Induced Pruritus," *Journal of Pediatric Pharmacy Practice*, 2000, 5(2):92-7.
126. Walley AY, Doe-Simkins M, Quinn E, Pierce C, Xuan Z, Ozonoff A. Opioid overdose prevention with intranasal naloxone among people who take methadone. *J Subst Abuse Treat*. 2013;44(2):241-247. [PubMed 22980450]
127. Wampler DA, Molina DK, McManus J, et al, "No Deaths Associated With Patient Refusal of Transport After Naloxone-Reversed Opioid Overdose," *Prehosp Emerg Care*, 2011, 15(3):320-4. [PubMed 21612385]
128. Weber JM, Tataris KL, Hoffman JD, et al, "Can Nebulized Naloxone Be Used Safely and Effectively by Emergency Medical Services for Suspected Opioid Overdose?" *Prehosp Emerg Care*, 2012, 16(2):289-92. [PubMed 22191727]
129. Weikert LF, Bernard GR. Pharmacotherapy of sepsis. *Clin Chest Med*. 1996;17(2):289-305. [PubMed 8792067]
130. Willman MW, Liss DB, Schwarz ES, Mullins ME. Do heroin overdose patients require observation after receiving naloxone? *Clin Toxicol (Phila)*. 2017;55(2):81-87. doi:10.1080/15563650.2016.1253846 [PubMed 27849133]
131. Wolfe TR, Bernstone T. Intranasal drug delivery: an alternative to intravenous administration in selected emergency cases. *J Emerg Nurs*. 2004;30(2):141-147. doi:10.1016/j.jen.2004.01.006 [PubMed 15039670]
132. Wong F, Edwards CJ, Jarrell DH, Patanwala AE. Comparison of lower-dose versus higher-dose intravenous naloxone on time to recurrence of opioid toxicity in the emergency department. *Clin Toxicol (Phila)*. 2019;57(1):19-24. doi:10.1080/15563650.2018.1490420 [PubMed 30032680]
133. Yousefifard M, Vazirizadeh-Mahabadi MH, Neishaboori AM, et al. Intranasal versus intramuscular/intravenous naloxone for pre-hospital opioid overdose: a systematic review and meta-analysis. *Adv J Emerg Med*. 2019;4(2):e27. doi:10.22114/ajem.v0i0.279 [PubMed 32322795]
134. Zimhi (naloxone) [prescribing information]. Carlsbad, CA: ZMI Pharma; August 2024.

Topic 9678 Version 645.0

Source: UpToDate. Downloaded 2026-05-26.